



# ASTITVA CHOUBEY



## ACADEMIC DETAILS

| Year | Degree / Board                           | Institute                               | GPA / Marks(%) |
|------|------------------------------------------|-----------------------------------------|----------------|
| ---  | M.Tech in Computer Science & Engineering | Indian Institute of Technology Delhi    | 7.27           |
| 2023 | B.Tech in Information Technology         | National Institute of Technology Raipur | 7.84           |
| 2019 | CBSE                                     | Krishna Public School, Nehru Nagar      | 88%            |
| 2017 | CBSE                                     | Krishna Public School, Nehru Nagar      | 10             |

## QUALIFYING EXAMS

- Graduate Aptitude Test in Engineering (GATE) Rank: 99

## IIT DELHI THESIS

**Title:** Large-Scale Recommender Systems Optimization at IndiaMART

**Supervisor:** Prof. Rahul Garg

**Description:** (IIT Delhi x IndiaMART) Engineered, implemented and evaluated E2E recommender pipelines on one of the largest e-commerce datasets in India. Led offline evaluation and live A/B testing to measure real business impact.

- Built a lightweight **frequency-based recommender** that consistently delivered **+34.6% CTR** in online A/B testing and experimented with a **SaaS recommender** that produced up to **75% increase in historical CTR** in offline evaluation.
- Researched, implemented and benchmarked matrix-factorization approaches (**NMF, IALS** and variants), and further proposed a **K-step Markov chain factorization** approach. Developed and tested both **gradient-descent** solver as well as **closed-form** solvers enabling the model to capture **sequential** and **latent** structure in user behavior.
- Tackled the challenge of **extreme data sparsity** by developing **product-clustering** strategies to aggregate similar SKUs, thereby improving the coverage of recommendations. In parallel, developed a **placement-bias model** to decouple **true product relevance** from its **position-induced bias** (quantifying how rank affects click probability).

## WORK EXPERIENCE

- **Morgan Stanley (Software Engineer):**

- Engineered and optimized Python **automation scripts** to intelligently **flag patterns** in business communications, improving **monitoring accuracy** and reducing **manual review effort**. Designed and executed comprehensive **end-to-end (E2E) test cases** to ensure **system reliability** and **data consistency**. Collaborated with global teams to drive the **integration of AI/ML-based anomaly detection systems** for monitoring internal and external communications, enhancing **risk detection** and **compliance** (Deployed by teamware)

## PROJECTS

- **Mini Browser (Prof. Smruti Ranjan Sarangi):** Architected and developed a **production-grade** web browser from the ground up, tackling core challenges in **parsing, rendering, and process isolation**. Implemented a **custom HTML parser** to construct the DOM/AST and **engineered a lightweight rendering engine** for page layout and visualization. Designed a **multi-threaded and multi-process architecture with inter-process communication (IPC)** to isolate tabs and ensure efficiency. Integrated **core browser features** like navigation controls, history management, and session restore, demonstrating **end-to-end system design, low-level programming, and performance optimization**.
- **Operating system extension (Prof. Smruti Ranjan Sarangi):** Engineered foundational operating systems features within the xv6 kernel. Developed **custom system calls** to enhance the shell with process history, dynamic call blocking, and file permission controls. Designed an **advanced process scheduler** with priority boosting and robust signal handling for sophisticated process management. Architected a **virtual memory system** from the ground up, featuring an adaptive page replacement policy and on-demand page swapping to efficiently manage limited physical memory.
- **Secure AV (Prof. Huzur Saran):** Designed a **multi-layered cybersecurity framework** for autonomous vehicles, mitigating threats like **GPS/LiDAR spoofing, CAN bus attacks, and V2X vulnerabilities**. Built **ML-based anomaly detection with 90%+ accuracy** and a unified monitoring dashboard. [Additional projects](#)

## TECHNICAL SKILLS

- [Codeforces](#) (1800+ rating) | **Languages:** C/C++, Python, Typescript, SQL | **Tools and Libraries:** Unity, CMake, GTest

## POSITIONS OF RESPONSIBILITY

- **Head Coordinator, Technocracy:** Directed the sponsorship domain for a flagship institute event, guiding sub-teams in **fundraising, financial planning, and expenditure management**. Strengthened the event's financial backbone by structuring and aligning multiple teams toward a **common vision**, honing key **leadership and organizational** skills.

## TEACHING ASSISTANTSHIP

- Digital logic and system design (**COL 215**) | Data Structure and Algorithm (**COL 106**)



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## IIT COURSE

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| M.Tech in Computer Science & Engineering | Indian Institute of Technology Delhi | 7.27 | ---        |

## COURSES DONE

Advanced Data Structures, Computer Networks, Intro. To Logic & Funct. Prog., Software Systems Laboratory, Resource Management In Computer Systems, Networks & System Security, Minor Project